

Array Statistics

Input file: **standard input**
Output file: **standard output**
Time limit: 1 second
Memory limit: 256 megabytes

Given an array of integers, compute the following statistics:

- **mean**: the average value of all elements.
- **median**: the middle element after sorting the array*.
- **min**: the minimum element.
- **max**: the maximum element.
- **mx_even**: the maximum even element.
- **mx_odd**: the maximum odd element.

If the array contains no even elements, print -1 for **mx_even**.

If the array contains no odd elements, print -1 for **mx_odd**.

after doing any calculation, if any floating point appears, just ignore it and print the integer value.

Input

first line of input contains a single integer n ($1 \leq n \leq 100$) the size of the array.

second line of input contains n integers $n_1, n_2, \dots, n_n$ ($1 \leq n_i \leq 10$) the array elements

Output

Print the required statistics in the order they appear in the statement, each on a new line.

Examples

standard input	standard output
5 1 2 3 4 4	2 3 1 4 4 3
5 6 8 10 12 14	10 10 6 14 14 -1

*After sorting the array, the median is the middle element. If the array has an even number of elements, the median is the average of the two middle elements.